M. Sc. Physics (Semester-II) 2026
Quantum Mechanics II – Tutorial 4

1. Recall that operator implementing finite rotation by an angle θ along $\hat{n}$ is given by $\hat{R}_{\hat{n}}(\theta) = e^{\frac{i}{\hbar}\hat{J}.\hat{n}\theta}$. For the case $j = \frac{1}{2}$ consider the rotation around the $z-$axis by an angle θ.

   (a) Write the expression for $\hat{R}_z(\theta)$ as an exponential of the relevant Pauli matrix

   (b) Evaluate the exponential by summing up the relevant power series using the result of question 2 of home work sheet (i.e, using the values of σ_x^2,σ_y^2, and σ_z^2).

2. (a) Find the condition on ℓ' and m' under which the matrix elements $\langle n'\ell'm'|z|n\ell m\rangle$ are non-vanishing using the Wigner-Eckart theorem. Here $|n\ell m\rangle$ are the joint eigenstates of the Hydrogen atom Hamiltonian H, the angular momentum operator L^2, and L_z

   (**Hint:** identify z as a component of a tensor operator and use Wigner-Eckart theorem. The condition on ℓ' and m' corresponds to the non-vanishing condition of the respective CG coefficients. You may consider writing z in terms of Spherical Harmonics $Y_\ell^m(\theta, \phi)$ to identify tensor operator indices ℓ and m.)

   (b) Similarly, find the non-vanishing matrix elements $\langle n'\ell'm'| \mp \frac{1}{\sqrt{2}}(x \pm iy)|n\ell m\rangle$.

   Remark: The electric dipole operator ($\mathbf{d} = -e\mathbf{r}$) is a rank-1 irreducible spherical tensor operator ($k = 1$) used to calculate transition matrix elements $\langle \psi_f|\mathbf{d}|\psi_i\rangle$. The Wigner-Eckart theorem simplifies these by factoring them into a geometrical Clebsch-Gordan coefficient and a physical, state-independent reduced matrix element, crucial for selection rules. To apply the Wigner-Eckart theorem, $\mathbf{d}$ is expressed as a spherical tensor operator of rank $k = 1$. Rather than evaluate a large number of transition integrals, the theorem allows us to evaluate a single reduced matrix element together with appropriate Clebsch Gordon coefficients which are non-vanishing only for angular momenta labels satisfying addition of angular momenta rules.